



Time Machine

Time Machine 



Easy

General Skills

picoCTF 2024

browser_webshell_solvable

git

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Hints 

Description

1 2 3

What was I last working on? I remember writing a note to help me remember...

You can download the challenge files here:

- [challenge.zip](#)

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Author: The Analyst: Hyposelenia

Challenge: Time Machine

Category: General Skills

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I. Objective

Identify what the author was last working on by analyzing the Git history of the provided files and retrieve the hidden flag.

II. Background

Git is a version control system that records changes made to files over time. Each saved change, called a ***commit***, can include an **optional message written by the developer to describe what was done or why the change was made**. These commit messages are often used as reminders when reviewing past work.

In cybersecurity and CTF challenges, sensitive information is sometimes accidentally exposed not inside files, but within Git history itself—especially in commit messages.

III. Tool Used

- **Oracle VirtualBox (Kali Linux)** – Used as the Linux environment
- **Linux Terminal** – Used to execute decoding commands
- **Git** – Version control system used in the challenge

IV. Methodology

1. Downloaded the zipped file provided.
2. Launched Oracle VirtualBox – Kali Linux as my personal medium to solve this challenge.
3. Unzipped the zipped file.



```
(kali@kali)-[/media/sf_Downloaded_Cybersecfiles]
$ unzip challenge.zip
Archive: challenge.zip
  creating: drop-in/
  inflating: drop-in/message.txt
  creating: drop-in/.git/
  creating: drop-in/.git/branches/
  inflating: drop-in/.git/description
  creating: drop-in/.git/hooks/
  inflating: drop-in/.git/hooks/applypatch-msg.sample
  inflating: drop-in/.git/hooks/commit-msg.sample
```

4. Change directory to the extracted zip file

```
(kali@kali)-[/media/sf_Downloaded_Cybersecfiles]
$ cd drop-in
(kali@kali)-[/media/sf_Downloaded_Cybersecfiles/drop-in]
```

5. Listed all files, including hidden ones, using: “ls -a”

```
(kali@kali)-[/media/sf_Downloaded_Cybersecfiles/drop-in]
$ ls -a
. .git message.txt
```

6. Noticed the presence of a “.git” directory, indicating that the folder is a Git repository.
7. Attempted to read the text file using cat, but found no flag inside the file contents.

```
(kali@kali)-[/media/sf_Downloaded_Cybersecfiles/drop-in]
$ cat message.txt
This is what I was working on, but I'd need to look at my commit history to know why ...
```

(Didn't actually do this part, but this is to show what it looked like. The hint told us it won't help (it gave a clue ig).)

8. Used the Git command:

git show message.txt

to inspect the commit where the file was last modified, I see it as check history.



9. Observed that the file had only been created once and never edited.
10. Identified that the flag was located inside the **Git commit message**, not the file itself.
11. Retrieved the flag.

V. Result

picoCTF{t1m3m@ch1n3_e8c98b3a}

```
(kali㉿kali)-[/media/sf_Downloaded_Cybersecfiles/drop-in]
$ git show message.txt
commit 712314f105348e295f8cadd7d7dc4e9fa871e9a2 (HEAD -> master)
Author: picoCTF <ops@picoctf.com>
Date: Tue Mar 12 00:07:26 2024 +0000

    picoCTF{t1m3m@ch1n3_e8c98b3a}

diff --git a/message.txt b/message.txt
new file mode 100644
index 0000000..4324621
--- /dev/null
+++ b/message.txt
@@ -0,0 +1 @@
+This is what I was working on, but I'd need to look at my commit history to know why...
\ No newline at end of file
```

VI. Explanation

In earlier challenges (such as *First Find* or *Big Zip*), the flag was hidden **inside files** and could be recovered using tools like `grep` or by navigating directories. However, this challenge works differently.

Here, the file `message.txt` contains a note stating that the author needs to check their commit history. This is a direct hint that the flag is **not in the file**, but somewhere in Git's history.

Git commit messages are short notes added when saving changes. They act like reminders for developers. In this challenge, the author accidentally stored the flag as a commit message when the file was created.

By using `git show message.txt`, Git displays:

- **The commit metadata**
- **The commit message**
- **The changes introduced in that commit**



Since the file was never edited after creation, the commit message becomes the only meaningful place where “what the author was last working on” is recorded—revealing the flag.

When committing changes in Git, developers often write messages like:

`git commit -m "Hyposelenia here, I modified the file tee-hee"`

That quoted text is the commit message. In real-world security incidents, secrets have been accidentally leaked this exact way.

VII. Conclusion

This challenge demonstrates that sensitive data is not always stored inside files. Sometimes, it exists in version control metadata such as commit messages. Understanding how to inspect Git history is an important skill in both CTFs and real-world cybersecurity investigations.

By recognizing the difference between file contents and commit metadata, the flag was successfully retrieved.

— The Analyst: Hyposelenia